

RAGHAV KHANDLWAL

206-953-1067 | raghav.k.0847@gmail.com | linkedin.com/in/raghav-k847 | github.com/Raghav847

Education

Washington State University

Master of Science in Computer Science

Dec 2025

Pullman, WA

- **Relevant Coursework:** Advanced Algorithms, Distributed Systems, Machine Learning, Database Systems

Experience

Community Dreams Foundation

Mar 2026 – Present

Software Developer

Remote, USA

- Architected a TypeScript/Express REST API for program staff, replacing 10+ hours/week of manual donor record exports with self-serve access across volunteer, staff, and admin permission tiers
- Integrated Stripe and Mailchimp APIs into a donor management backend, automating recurring payment processing and email campaign triggers across client-facing workflows
- Implemented JWT-based RBAC with layered Express middleware guards, enforcing permission boundaries across volunteer, staff, and admin tiers for auditable access control
- Built an async background job system with Redis and Bull for report generation and email dispatch, decoupling long-running tasks from the request cycle and cutting client-facing UI wait time by 70%

Headstarter AI

Jun 2024 – Jul 2024

Software Engineering Fellow

Remote, USA

- Deployed a full-stack, JWT-secured LLM chatbot (Next.js, OpenAI streaming, WebSockets) sustaining sub-300ms latency across 500+ concurrent sessions

DXC Technology

Jun 2022 – Nov 2022

Associate Professional Software Engineer

India

- Designed and deployed a Python and SQL ETL pipeline extracting, transforming, and loading 6+ GB/month from DB2 into PostgreSQL, resolving integration failures between legacy and modern database environments
- Developed RESTful APIs for insurance policy operations, sustaining 99.5% uptime at 500+ RPM with documented endpoint contracts for downstream consumers
- Containerized 4+ enterprise applications with Docker on AWS (EC2, RDS, S3), reducing environment setup time by 60% and enabling consistent deployments across dev, staging, and production
- Optimized SQL Server and DB2 query performance via index tuning and join refactoring, reducing response latency from 4.2s to 800ms

Projects

Greenlight – Go REST API | Go, PostgreSQL, net/http

[GitHub](#)

- Engineered a production-grade Go REST API with PostgreSQL, implementing optimistic concurrency control via version-based conflict detection to prevent lost updates under concurrent writes
- Implemented token-based authentication and per-client rate limiting middleware, enforcing secure session management and protecting endpoints from abuse
- Built full-text search over PostgreSQL using GIN-indexed columns, enabling efficient filtered queries across large result sets

Snowpack Prediction – AgAid Hackathon | Python, PyTorch, Flask, React

[Link](#)

- Designed a Transformer-based neural network for Snow Water Equivalent prediction, achieving 87.8% accuracy across 9M+ spatio-temporal records sourced from 8 meteorological datasets
- Deployed a Flask and React prediction service with real-time CSV ingestion and interactive SWE visualizations

AI Agent | TypeScript, Node.js, Python, OpenAI API, RAG, PyTest

[GitHub](#)

- Engineered a multi-tool AI agent with a RAG pipeline, vector embeddings, and OpenAI API integration across TypeScript and Python services
- Improved output relevance by 30% via semantic retrieval and reusable prompt configurations across varied content workflows

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, SQL

Frameworks & Libraries: Node.js, Express.js, Next.js, React, Flask, PyTorch

Backend & Systems: REST APIs, JWT, RBAC, Rate Limiting, Optimistic Concurrency Control, Full-Text Search, Redis

APIs & Integration: WebSockets, OpenAI API, Stripe API, Mailchimp API

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Git

Databases: PostgreSQL, DB2